

Heard but Not Studied: Mapping the Gap Between Health Research and Women's Lived Experiences

Data 510 Project Proposal

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1. Title and Team Metadata

Project Title: Heard but Not Studied: Mapping the Gap Between Health Research and Women's Lived Experiences

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Studio Session: 3

Team Size: 2

2. Problem Framing and Research Questions

2.1 Background

Women represent just over half the global population, yet for decades they have been systematically underrepresented in biomedical research. Historically, men's bodies were treated as the default in medical study design,^[1] with findings generalized to women despite important biological differences. Most of the time, this was not accidental; researchers often actively avoided including female subjects due to concerns about hormonal variability interfering with results, the cost of housing female animals, and even because of policies designed to shield women of reproductive age from experimental drug exposure.^[1] The consequences have been serious, where women experience adverse drug reactions at twice the rate of men,^[2] wait longer for both diagnosis and pain relief,^[3] and are more likely to be misdiagnosed during serious

medical events.^[3] This is a pattern so well documented that it has been given a name, Yentl Syndrome, which is the phenomenon where a woman's illness must first mirror a man's presentation before it is taken seriously.^[3]

Legislative intervention has tried to correct this imbalance. In 1993, the U.S. Congress required the inclusion of women in federally funded clinical trials^[1], and the 2016 Twenty-First Century Cures Act went further by mandating that trial results be stratified by sex, gender, race, and ethnicity.^[1] Despite this progress, gaps remain substantial. Studies have been done to evaluate the National Institutes of Health (NIH) funding patterns. These studies have exposed the persistent disparities favoring diseases that predominantly affect men.^[4] Conditions like endometriosis, uterine fibroids, autoimmune disorders, and Polycystic Ovarian Syndrome (PCOS) remain underfunded and understudied relative to their prevalence.^[4] As of May 12, 2026, PMOS (Polyendocrine Metabolic Ovarian Syndrome) is the current term for PCOS to more accurately reflect the condition and its presentation; however, since most studies and communities still refer to it as PCOS, we will use PCOS for consistency. Endometriosis is estimated to affect around 10% of women worldwide, yet diagnosis can take up to 10 years,^[6] in part because endometriosis is frequently dismissed as psychological.

Gender disparity in research is hard to measure because it involves the “absence of research”, which is difficult to see in literature. There have been many studies that have made progress on quantifying the gap from participant data.^[8] For example, a 2019 study quantified sex bias in clinical studies using automated data extraction to assess the magnitude of female underrepresentation. They found substantial underrepresentation of women in 7 of the 11 disease categories they analyzed.^[8] While these studies are extremely important, their approaches rely on what has already been studied and published, so they cannot analyze what women are actually experiencing.

Online health communities have emerged as a large source of what burdens women experience in healthcare. Platforms like Reddit host active communities for conditions such as r/PCOS, r/endo, r/Fibrosis, r/periods, r/vaginisums, r/WomensHealth, and so much more. Women can share symptoms, treatments, and healthcare experiences. These discussions are extremely important, as they reflect the lived experiences of women that are often missing from published clinical research. Treating these spaces as data offers a way to locate that gap, or “missingness”, between what women experience versus what the research community prioritizes. There is new research on these kinds of spaces, including a 2025 study^[11] analyzing online communities formed around menopause. Their main research angle was the quality of information in these communities, examined through thematic analysis. However, there is very little research comparing women's lived experiences with published health literature. Our project aims to help close this gap.

2.2 Problem Statement

Problem: Women have been historically underrepresented and underprioritized within medical and clinical published research, leading to a negative impact on women's ability to understand their own health issues, receive diagnosis, treatment, and easily navigate healthcare systems.

Consequence if not solved: If the gender gap disparity in published health research is not closed, health systems will continue to burden women with self-advocacy in spaces that have not been designed to support them. Women will remain understudied, underfunded, and misunderstood.

What success would change? There will be a fully mapped understanding of what women discuss online, where published health research is missing, which topics are understudied and which are not studied at all, and what should be prioritized in future published research.

2.3 Research Questions

RQ1: What women's health symptoms and conditions are most discussed in online communities, and to what extent are these topics represented in clinical research?

Backup RQ1 or RQ2:

Which women's health conditions show the largest disparity between their documented disease burden (GBD data) and their representation in published clinical trial enrollment on [ClinicalTrials.gov](https://clinicaltrials.gov)?

3. Impact and Stakeholders

Stakeholders and Impact: The primary stakeholder for this project is women. Stakeholders who will make that happen are medical researchers, clinical trial designers, and public health policymakers who decide where to allocate research funding and direct study enrollment. If this project succeeds, it will produce a data-driven, ranked map of women's health conditions that are widely discussed in patient communities but underrepresented in clinical literature, giving decision-makers a concrete, evidence-based starting point for prioritizing understudied conditions. Secondary stakeholders include patient advocacy organizations (such as endometriosis or PCOS foundations) who can use gap findings to strengthen funding arguments, and then women patients themselves, who stand to benefit downstream if research priorities shift toward conditions they disproportionately experience.

Shared Theme and Peer PO Influence: Our shared theme with our peer POs is *health and identity through text-based disparity analysis*.

From this perspective, our peer reviewers will likely hold us accountable to responsible text analysis practices, including how we handle anonymity, consent, and the ethics of scraping health-related personal narratives from public forums.

4. Data Sources and Access Plan

Data Source	Description	Access Plan	Estimated Volume	Refresh	Instructor approval
Women's Health Dashboard https://swhr.org/programs/women-s-health-dashboard/ (CONTACT)	This is the Women's Health Dashboard from the Society for Women's Health Research. Their aim is to offer a platform to explore national and state data on diseases and health conditions that have significant impacts on women's health. This includes public health data, research investment and progress, health insurance coverage, and relevant policies.	We would need to contact development@swhr.org or info@swhr.org to get access to the data because it is no longer being updated as of 2026.	Unsure	N/A	Approved ▾
ClinicalTrials.gov More specifically: https://clinicaltrials.gov/data-api/api	This is an online database that includes information about clinical research studies and information about their results.	We would access this data through their OpenAPI. Data is refreshed daily.	500,000+ studies, the full dataset is quite large.	One time pull.	Approved ▾
NIH RePORTER	This data source is from the National Institutes of Health (NIH) and includes	Access to this data would be through their API, the NIH	Data from 1972 to 2012 in historical	This would be a one-time data pull	Approved ▾

https://reporter.nih.gov/	monthly grant awards, grant values, funding by state in the US, active projects by institute/center, and advanced projects with information on year, PI, organization, agency, and project number.	RePORTER APIs.	records. Currently, over 500,000-600,000 active and recently closed projects.	from the API.	
PubMed APIs https://pmc.ncbi.nlm.nih.gov/tools/developers/	NCBI's database contains biomedical literature. The API is called E-utilities and allows searches for articles, full record details, and related records.	We would access the data using APIs and the base URL for whichever API. There are 5 different APIs available.	PubMed has over 40 million citations starting back in 1966. They add over 1 million articles each year. The size would depend on how much data we want to work with.	This would most likely be a one-time pull from the APIs.	Approved ▾
Global Burden of Disease https://www.healthdata.org/research-analysis/gbd Population-level health statistics.	The GBD is the largest and most comprehensive effort to quantify health loss across places and over time. It looks at 204 countries and territories and 463 different health outcomes and risk factors.	Use the GBD results tool to download up to 100,000 rows. For more rows, submit a request form .	One-time download	N/A	Approved ▾

ScienceDirect https://www.sciencedirect.com/ (CONTACT)	This is one of the most popular places for researching peer-reviewed journal articles and books. *It will be very hard to get an API; full access is normally granted to researchers affiliated with organizations (so maybe we could?) Institutional access: full text and metadata Without institutional access: metadata and abstracts API is limited to 200 results per query.	Access might be hard; we need to request an API key, but make sure it is valid through the University.	Smaller	Might have to pull multiple times, due to how much data we can get from the API.	Approved ▾
Nature Medicine https://www.nature.com/nm/ https://dev.springernature.com/	A monthly, peer-reviewed medical journal for translational and clinical research. It is one of the most prestigious medical journals in the world, published by Springer Nature. They have a Meta API and an Open Access API. I believe we would need to subscribe.	Access APIs	15,000-25,000 articles	One-time pull.	Approved ▾
paperscraper	Python package for web scraping metadata or	Free- install the Python package and use it	30,000 - 400,000 papers	Depends on how many places we	Approved ▾

https://pypi.org/project/paperscraper/	full-text files from PubMed and servers: arXiv, medRxiv, bioRxiv, and chemRxiv. This will help us gain a deeper understanding beyond what PubMed's API will give us and give us counts for citations, like in Google Scholar			scrape from.	
PRAW library (Python Reddit API Wrapper) for the Reddit API https://developers.reddit.com/docs/capabilities/server/reddit-api	This will help to support retrieving Reddit post data. It can scrape posts, comments, subreddits, users (data we will not use), and searches.	Free for non-commercial research use, though rate-limited	Around 1,000 posts.	Maybe 2 or 3 pulls, depending on the size of the data we want to pull.	Approved ▾
General scraping tools like BeautifulSoup and Scrapy for online forums https://beautiful-soup-4.readthedocs.io/en/latest/ https://github.com/scrapy/scrapy	Beautiful Soup: Python package, HTML and XML parser. Scrapy: A web-scraping tool to extract structured data.	Available as a Python package. Make sure to check the websites' robots.txt before using	Around 1,000 posts.	Again, maybe 2 or 3 pulls.	Approved ▾

5. Data Engineering Plan

We will use Railway to host a PostgreSQL database for our data storage. We decided to use a separate database and not the GitHub repository to ease storage constraints and privacy concerns. Data will be organized into three layers: raw (original downloads/scrapes; never overwritten; used as staging tables in Railway), interim (harmonized/joined tables), and processed (analysis-ready). In the GitHub repository, data/ will contain documentation on the data collection process, issues we ran into, how we resolved them, schema documentation, and any other notes.

Ingestion scripts will live in src/ingest/. For sources with stable bulk downloads (one-time CSV or API pulls), we will download locally, clean, and upload to a staging table in Railway, including any code in notebooks/ or src/ingest. For scraping sources that are rate-limited and need to be run at intervals, we will implement a scheduled scrape that feeds directly into Railway, into a predefined staging table. All transformation steps from raw to interim to processed will be documented in data/ with references to specific scripts.

We will resolve the one-time vs. pipeline scraping decision by Milestone 2, based on the sources we use and their rate limits. We expect to have data created by the modeling we will do, so we will have another layer of data in Railway: outputs, where src/models/ will write to this part of the database.

6. Planned Analysis

Perform statistical analysis on any numerical data we obtain, but our data will consist mostly of text data. We will use text analysis tools like topic modeling, named entity recognition, frequency analysis, and sentiment analysis. We will do a statistical analysis on the results of the text analysis. Potential tests and tools. BERTopic:

<https://maartengr.github.io/BERTopic/index.html> spaCy: <https://spacy.io/>. Statistical Tests: Chi-test, ANOVA, T-test, and/or Pearson Correlation.

Example PBIs:

PBI	Create/Observe/Analyze
PBI1: Topic Modeling on published articles from PubMed	Create: Train BERTopic model on cleaned API article data. Observe: Generate different topic groups with keywords. Analyze: Evaluate different topics and how they relate

	to domains. Potentially apply a score that maps between frequency of topic in published data and forum data.
PBI2: Gap Analysis	Create: Build a way to compare between forum topics and research topic volume. Observe: Visualize the comparison Analyze: Use chi-squared test to understand how statistically significant the difference is within different topics.

7. Program Integration

- 1. Data Engineering:** We will scrape data from raw sources, clean and transform it into usable datasets, and store it in PostgreSQL. This uses a basic ETL pipeline common in data engineering. The database will be maintained with the ultimate goal of using the data for analysis.
- 2. Analytics and Machine Learning:** We will conduct statistical analysis of numerical data we collect or generate through text analysis. The text analysis we will conduct may include topic modeling, named entity recognition, frequency analysis, and sentiment analysis.
- 3. Visualization:** Exploratory visualizations will be created, and final visualizations will be shown in the poster, write-up, and dashboards, if applicable.
- 4. Ethics:** We plan to use data for a good cause, but if we do so incorrectly, we can be misleading and divert attention from important topics that need further study. Another ethical consideration is how we will process, store, and use the sensitive data we collect. We will aim to anonymize data to the best of our ability, omitting personal information from the text or removing the entire data entry if needed. We will carefully analyze the outputs of any modeling we do, ensuring no sensitive data is produced. We also need to acknowledge that women go to these forums to seek help from other women, but most of the women there are anonymous. They are not consenting to their data being used.
- 5. Statistics and Research Design:** Most of our analysis will consist of text-based analysis, comparing between forum and published data. We will use machine learning methods like topic modeling, named entity recognition, frequency analysis, and sentiment analysis. Depending on the variables we can collect and scrape, we will also conduct statistical analyses, including the chi-squared test, Pearson correlation, ANOVA, and t-tests.

8. Ethics and Risk Notes

Ethics and Risk Notes: Privacy, consent, fairness, retention, and known limitations of the data or setting. These should match the *Ethics risks* bullets in your [CHARTER.md](#) Context section.

- We plan to use data for a good cause, but if we do it incorrectly, we can be misleading and divert attention from important topics that need further study.
- Another ethical consideration is how we will process, store, and use the sensitive data we collect. We will aim to anonymize data to the best of our ability, omitting personal information from the text or removing the entire data entry if needed. We will carefully analyze the outputs of any modeling we do, ensuring no sensitive data is produced.
- We are limited to a few data sources for both research and discussion. Getting a general sense of “what is being researched” and “what is discussed most” is difficult and is limited to the data sources we access. In our findings, we will clarify the exact scope of our assessment.
- Scraping is another ethical issue- we have to ensure we have permission from websites to scrape their content. Scraping research papers is another issue, as it borders on pirating and can violate publisher permissions.
- It is also important to note the disparity in research on the intersectionality of race and gender, which we are not covering in our analysis. Women of color face systemic components, including stereotyping, insurance access, and accessible healthcare, that all contribute to the imbalance of equity for racial groups within healthcare. Medical literature overall struggles to report data regarding sex and gender; it struggles even more so with reporting race and ethnicity.^[7] This ethical issue will be noted in our analysis and data collection, specifically when gathering data from online forums, although it is not the focus of our research.

9. Timeline and Milestones

Week	Plan
4 (Now)	This week, we will start collecting data. This will involve accessing APIs for online health journals and web scraping online forums. Once we have started collecting our data, we will convert it into dataframes, perform transformations, and clean it. These datasets will be stored in Postgres/Railway.
5	This week, we will continue collecting data. Then, transforming it to put into our pipeline. Once we have collected all our data sets, we will start running analytical queries and exploring relationships.

6	By this week, we aim to have collected all of our data. Regardless, our pipeline should be solidified. Datasets should connect, and we will start the analysis in Python. We will do text analysis in Python, including topic modeling, named entity recognition, and frequency analysis. Depending on our transformations and the data we can collect, we will conduct comparative statistical tests such as t-tests, ANOVAs, chi-tests, or Pearson correlations.
7 M2-data-summary	This week, we will have our data summary complete. This will include our completed data collection, full ERD, data pipeline, and table storage. We will ensure that data quality checks are completed and that our transformations are complete.
8	This week, we will continue with statistical tests, modeling in Python, and analysis. Hopefully, by this time, we will have a solid idea of what our final dashboard will look like.
9	Continue modeling and statistical tests. By this week, we should start working on our final dashboard, knowing what information we will display based on our initial research question.
10 M3-poster-draft	By this week, we should have our poster draft. This will be a week we want to be almost fully done with statistical testing and modeling. We will want to have a coherent story and impactful results.
11	This week would most likely consist of writing out a draft. We will continue working on our dashboard and poster.
12 M4-write-up-draft	This week, our draft write-up is due. We will continue working on finalizing the paper, dashboard, and poster. By this time, we should feel confident we can complete everything by next week.
13	This week, we will finalize our write-up, dashboard, and poster. Make sure everything is ready and practice presenting.
14 M5-final	This week, everything is due. Make sure we are ready to present, the dashboard is up and working, the poster is ready, and the paper is done!

Risks and Mitigations:

Some risks include issues with data collection. One is scraping taking longer than anticipated due to rate-limiting, which we will mitigate by scheduling scraping to run overnight for long periods of time. Another risk is unexpected access issues, which we mitigate by maintaining multiple data sources as backup options. Transforming and storing the data can be computationally expensive, so we will use Railway to host our database and upgrade our plan if necessary.

Modeling and analysis can be computationally expensive as well, so we will use Google Colab and its hosted runtimes to improve efficiency.

10. D3 Process and Tooling

- **GitHub repo URL** <https://github.com/Addie-Shanti-Org/GenderDisparityinResearch>
- **GitHub Projects (v2) board URL** <https://github.com/orgs/Addie-Shanti-Org/projects/3/views/1>
- **Discord category:** Our channel, Project 2: Addison & Shanti is confirmed, each channel is up and running, peers have joined, and we are opted into our own project's category.
- **CHARTER.md:** Charter is committed. **Vision:** If this project succeeds, we will help improve women's health by helping to eliminate gender gaps in research. **Mission:** Our mission is to identify gaps between the health issues women actually experience and what has been researched. We will do this by gathering data from online communities about what is being discussed and comparing it to what research has been done on women's health. Context matches proposal. Working agreements filled out.
- Confirmation that **CHARTER.md** is committed (vision, mission, context, success criteria, working agreements, **Response SLAs**, DoR/DoD) and that the backlog is seeded with **milestone-tagged PBIs**.
- **Studio Brief / Studio Critique:** Studio Briefs would be due around the time of each milestone, we will communicate dates directly with our peer stakeholders on discord and email. Studio Critiques will be due a varying number of days before the next milestone. We will communicate dates directly with our peer stakeholders on discord and email. Owner team integrates briefs in good faith; the instructor arbitrates (as Process Expert) if peer POs and owner team disagree.

11. Appendix

[1] AAMC. (2024). *Why we know so little about women's health*. Association of American Medical Colleges. <https://www.aamc.org/news/why-we-know-so-little-about-women-s-health>

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[3] Merone, L., Tsey, K., Russell, D., & Nagle, C. (2022). Sex Inequalities in Medical Research: A Systematic Scoping Review of the Literature. *Women's health reports (New Rochelle, N.Y.)*, 3(1), 49–59. <https://doi.org/10.1089/whr.2021.0083>

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[6] De Corte P, Klinghardt M, von Stockum S, Heinemann K. Time to Diagnose Endometriosis: Current Status, Challenges and Regional Characteristics-A Systematic Literature Review. *BJOG*. 2025 Jan;132(2):118-130. doi: 10.1111/1471-0528.17973. Epub 2024 Oct 7. Erratum in: *BJOG*. 2025 Jun;132(7):1018. doi: 10.1111/1471-0528.18149. PMID: 39373298; PMCID: PMC11625652

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[10] Khan S, Eshaghi F, Rehman MZ, Kotwal S, Syed M, Khan K and Kumar KS (2024) Are females in classical hematology getting a fair share? Uncovering gender disparities in NIH R01 grants. *Front. Sociol*. 9:1430369. doi: 10.3389/fsoc.2024.1430369

[11] Funnell EL, McConnell F, Martin-Key NA, Qian L, Babbitt K, Bahn S. Understanding Experiences of and Unmet Needs in Online Searches for Menopause Information: An Exploratory Survey. *JMIR Form Res*. 2025 Oct 1;9:e75335. doi: 10.2196/75335. PMID: 41032866; PMCID: PMC12530155